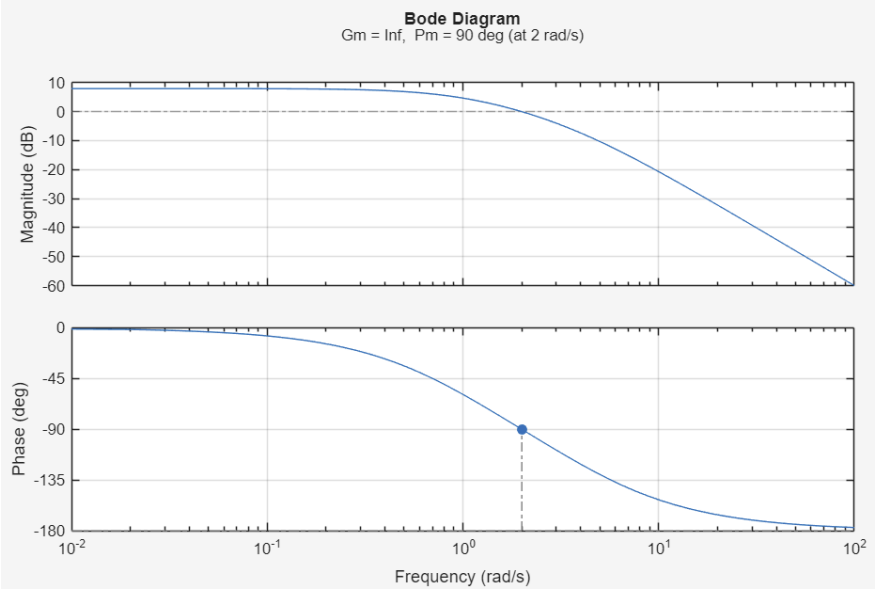
Problem 8.

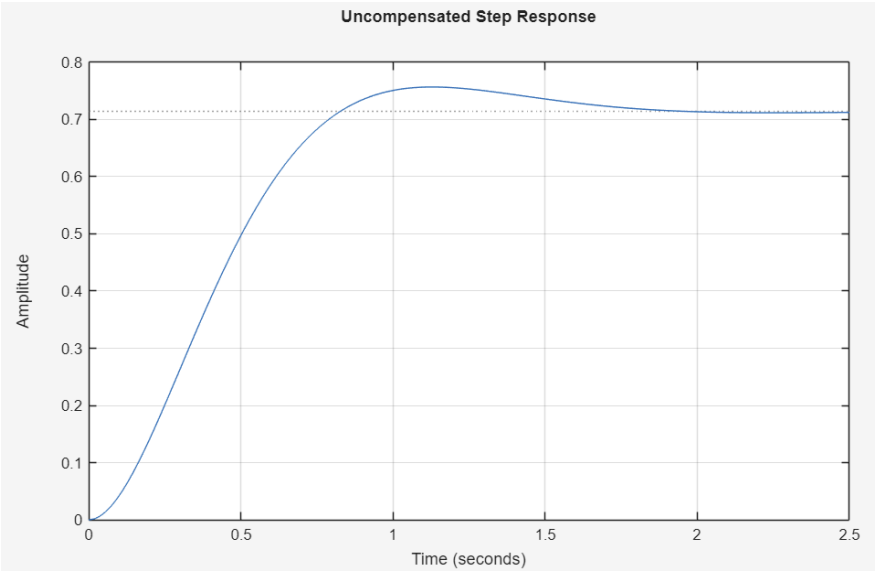
Add compensation to improve the closed loop behavior (both transient and steady-state response). Describe your process, the compensation selected and how you improved the response compared to the uncompensated system.

Understand the uncompensated system



The uncompensated system has a gain crossover frequency of $\omega_{gc} = 2$ rad/s and a phase margin of 90 degrees.

I can add a lead compensator to maintain a decent phase margin while improving system response speed by increasing gain K_c. I will arbitrarily decide on a phase margin of 45 degrees.



From MATLAB, the step-response time specs are as follows:

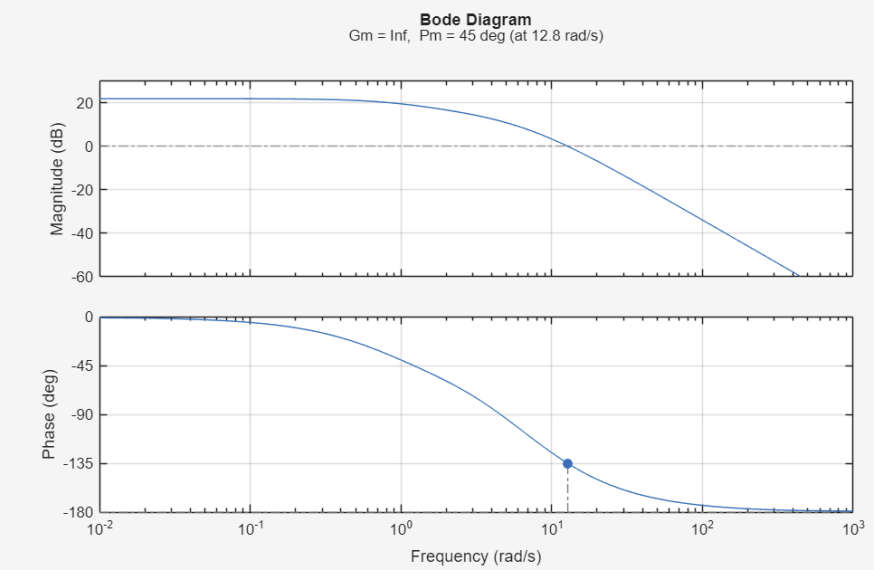
```
      RiseTime: 0.5435
 TransientTime: 1.6066
  SettlingTime: 1.6066
   SettlingMin: 0.6454
   SettlingMax: 0.7568
     Overshoot: 5.9523
    Undershoot: 0
          Peak: 0.7568
      PeakTime: 1.1237
```

```
Steady-state error: 0.2857
```

Since there is a steady-state error, I can add a lag compensator to reduce the steady-state error to a unit step input.

1 **Determine lead compensator parameters:** I will choose an arbitrary lead compensator zero and pole location of $z = 2$ and $p = 8$, such that $z < p$.

In MATLAB, I will iteratively increase the gain K_c until the phase margin is 45 degrees, aiming for a higher crossover frequency of about double the uncompensated system's crossover frequency ($\omega_{gc,\text{ desired}} \approx 4$ rad/s).



For $K_c = 20$, I found that

```
Gain crossover: 12.78 rad/s
Phase margin: 45.02 degrees
```

This is satisfactory, so I will keep the lead compensator parameters as $K_c = 20$, $z = 2$, and $p = 8$.

2 **Determine lag compensator parameters:** Solving the steady-state error of the lead compensated system:

$$L(s) = \frac{20(s+2)}{s+8} \cdot \frac{10}{(s+1)(s+4)} = \frac{200(s+2)}{(s+8)(s+1)(s+4)}$$

For a unit step input, the steady-state error is:

$$\begin{aligned} e_{ss} &= \lim_{s\to 0} sE(s) \\ &= \lim_{s\to 0} s \cdot \frac{1}{s} \cdot \frac{1}{1+L(s)} \\ &= \lim_{s\to 0} \frac{(s+8)(s+1)(s+4)}{(s+8)(s+1)(s+4)+200(s+2)} \\ &= \frac{32}{32+400} \\ &= \boxed{\frac{2}{27} \approx 0.0741}. \end{aligned}$$

Thus, I will need to add a lag compensator to limit the steady-state error to a unit step input to be less than or equal to 0.01. Finding the ratio z/p for the lag compensator:

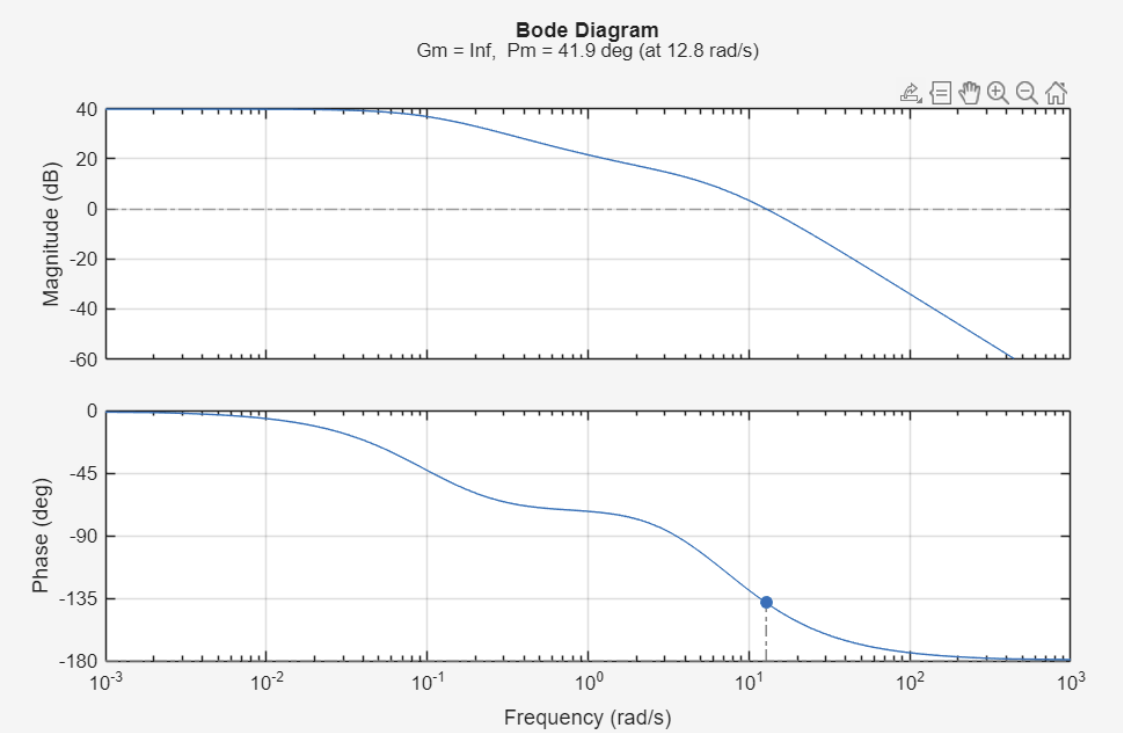
$$C_{lag} = \frac{s+z}{s+p}$$

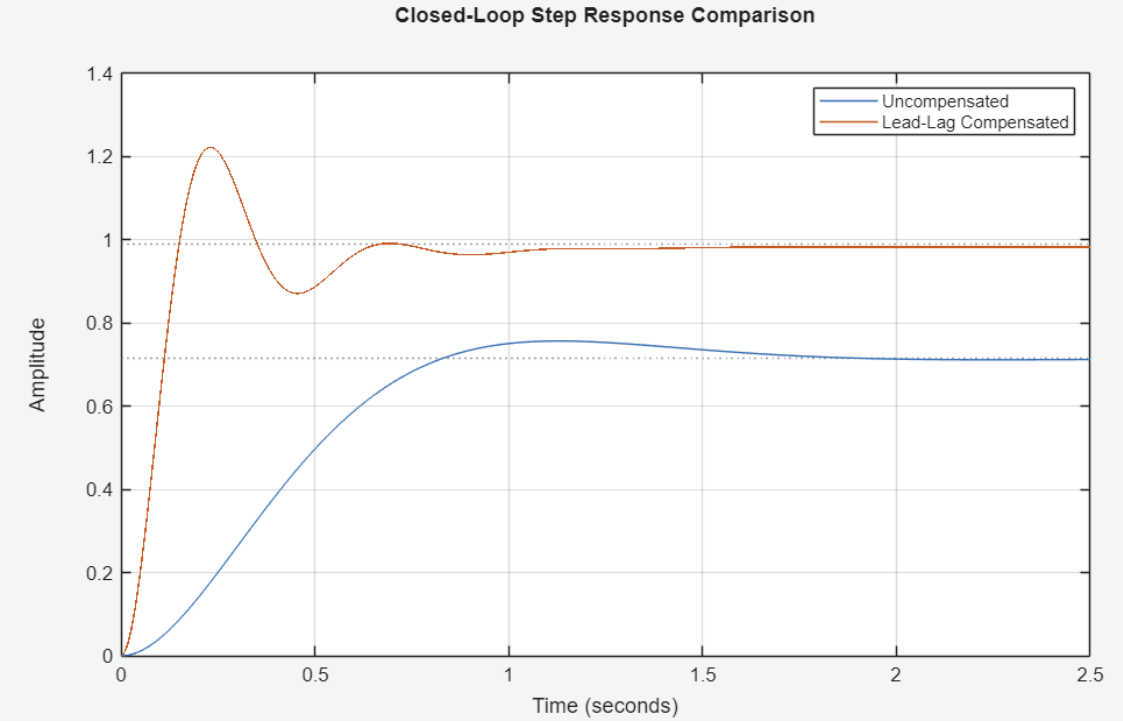
$$\begin{aligned} e_{ss} &= \lim_{s\to 0} sE(s) \\ &= \lim_{s\to 0} s \cdot \frac{1}{s} \cdot \frac{1}{1+L(s)} \\ &= \lim_{s\to 0} \frac{(s+8)(s+1)(s+4)(s-p)}{(s+8)(s+1)(s+4)(s-p)+200(s+2)(s-z)} \\ &= \frac{-32p}{-32p+400z} \\ &= \frac{32p}{32p+400z} \end{aligned}$$

$$\begin{aligned} e_{ss} = \frac{32p}{32p+400z} &\leq 0.01 \\ 32p &\leq 0.32p + 4z \\ 31.68p &\leq 4z \\ \frac{z}{p} &\geq 7.92 \end{aligned}$$

To keep the lag compensator's poles and zeros close to the origin, pick $p = 0.1$ and $z = 0.792$.

Final Compensated Response





Time specs:

```
    Lead-Lag Compensated specs:
Gain crossover: 12.7911 rad/s
Phase margin: 41.88 degrees
```

```
ans =

  struct with fields:

         RiseTime: 0.1013
    TransientTime: 1.0000
     SettlingTime: 1.0000
      SettlingMin: 0.8712
      SettlingMax: 1.2229
        Overshoot: 23.5227
       Undershoot: 0
             Peak: 1.2229
         PeakTime: 0.2299

Steady-state error: 0.0100
```